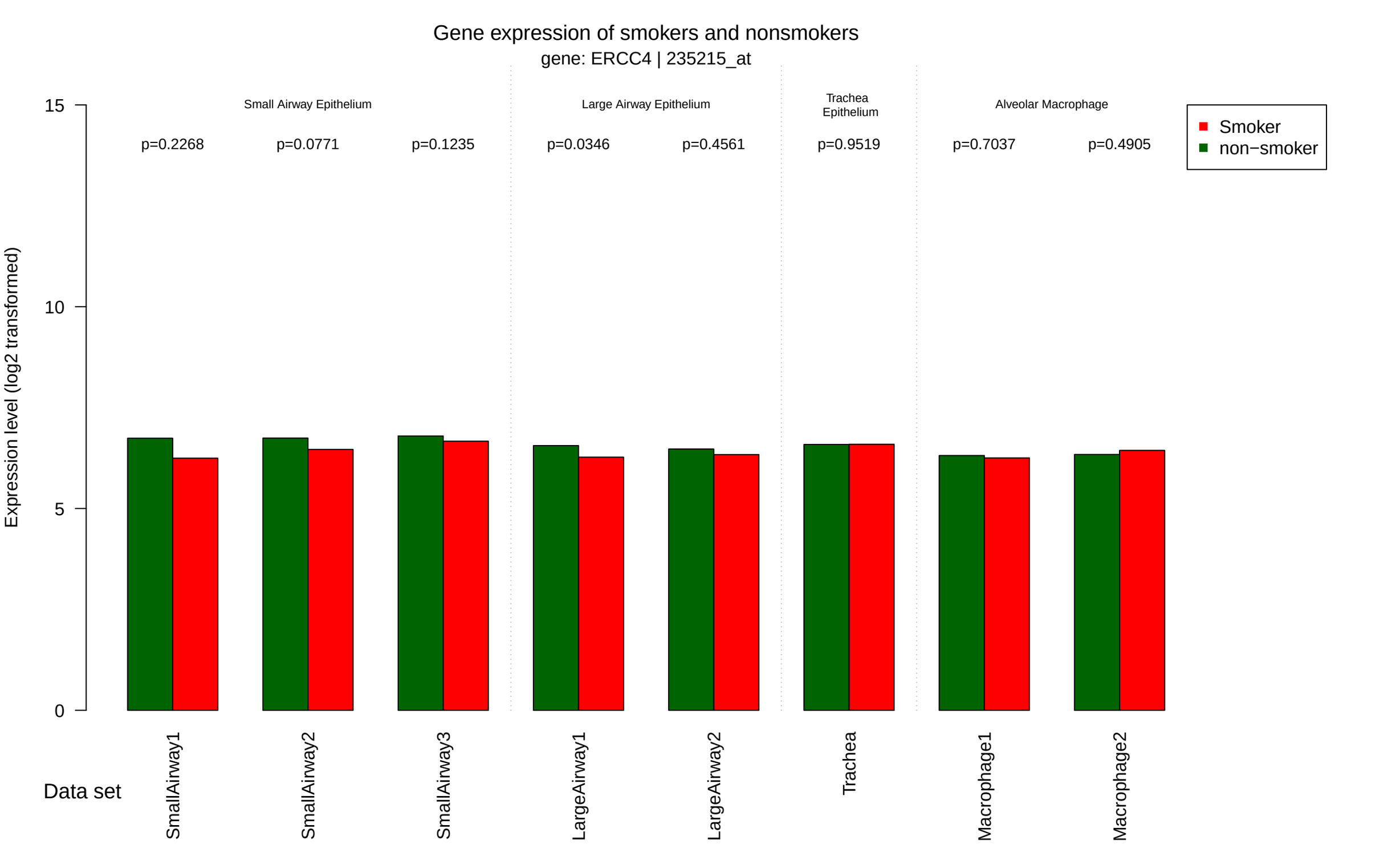
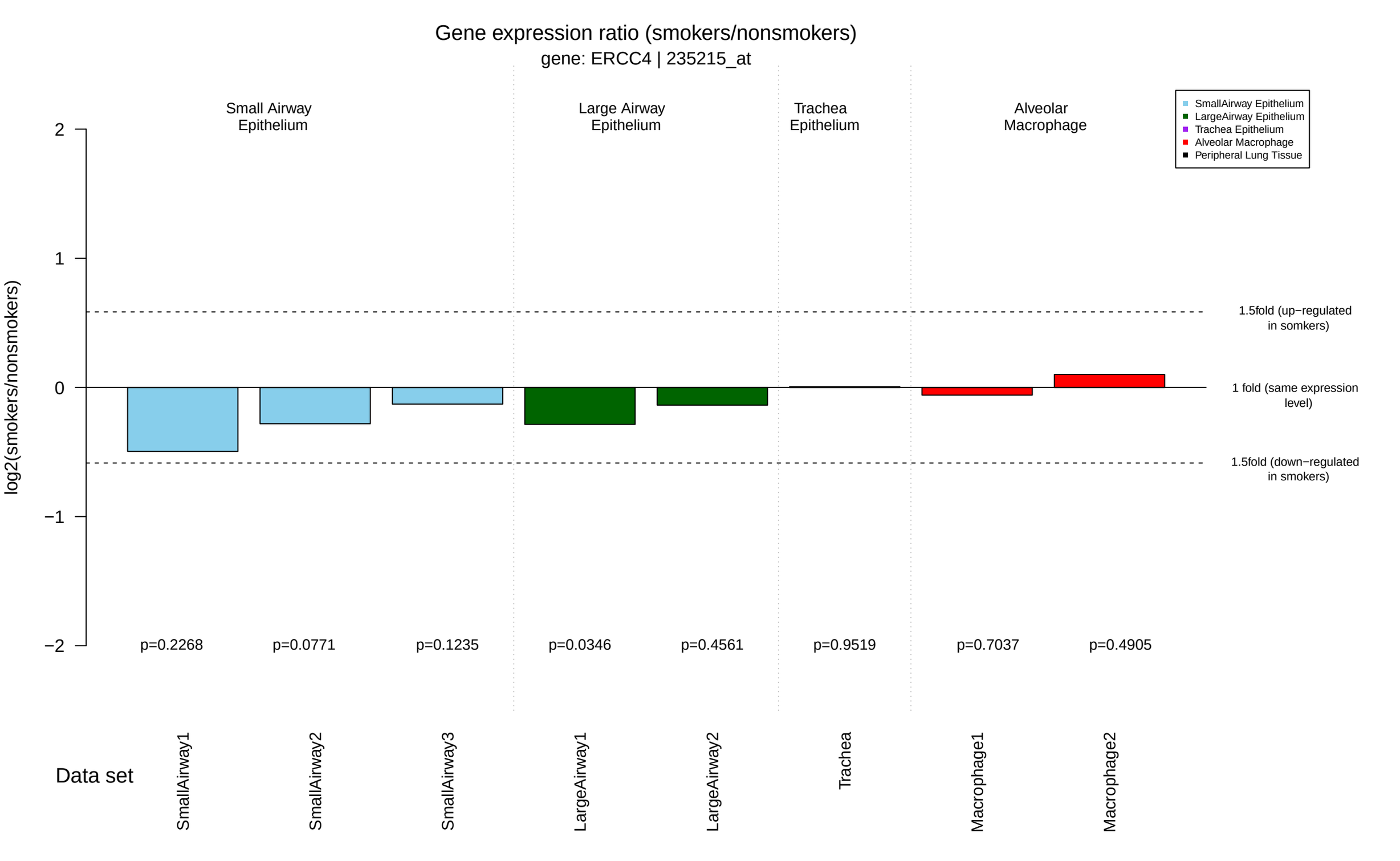
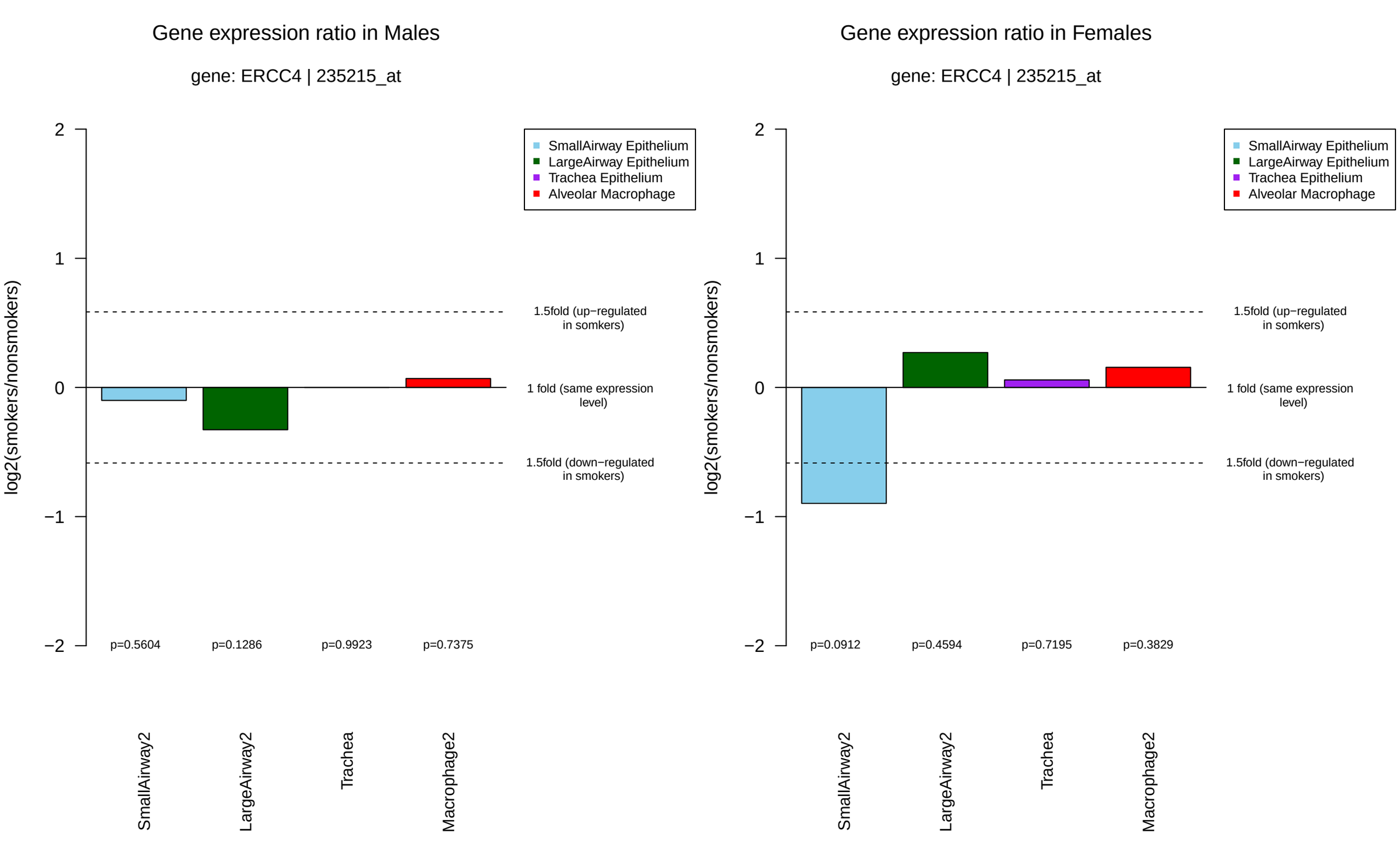
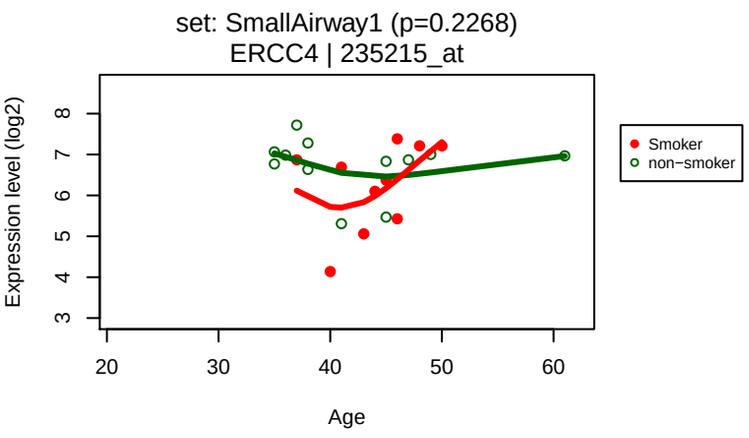
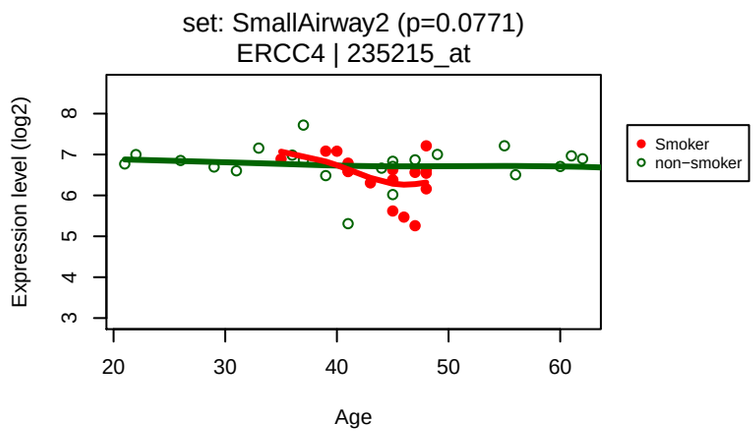
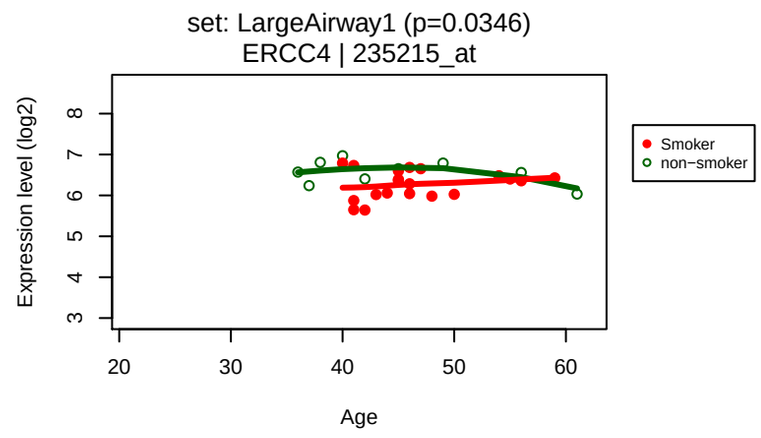

Gene expression of smokers and nonsmokers
gene: ERCC4 | 235215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2268
p=0.0771
p=0.1235
p=0.0346
p=0.4561
p=0.9519
p=0.7037
p=0.4905
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

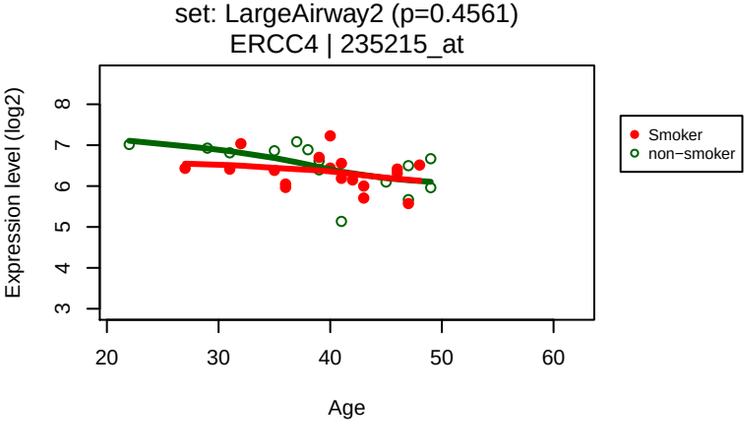

Gene expression ratio (smokers/nonsmokers)
gene: ERCC4 | 235215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2268
p=0.0771
p=0.1235
p=0.0346
p=0.4561
p=0.9519
p=0.7037
p=0.4905
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

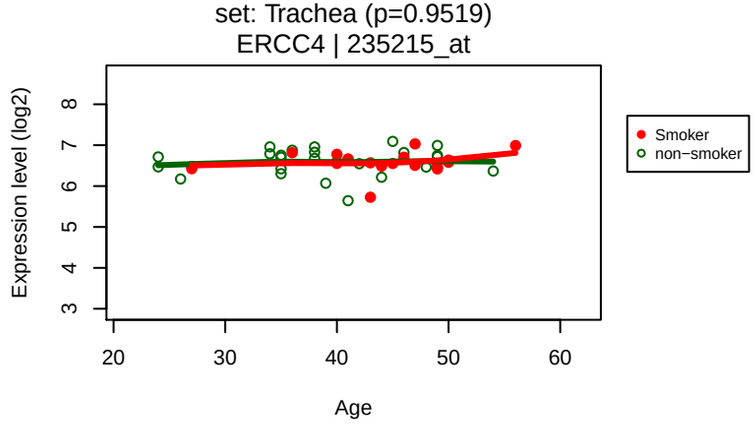

Gene expression ratio in Males
gene: ERCC4 | 235215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5604
p=0.1286
p=0.9923
p=0.7375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERCC4 | 235215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0912
p=0.4594
p=0.7195
p=0.3829
SmallAirway2
LargeAirway2
Trachea
Macrophage2

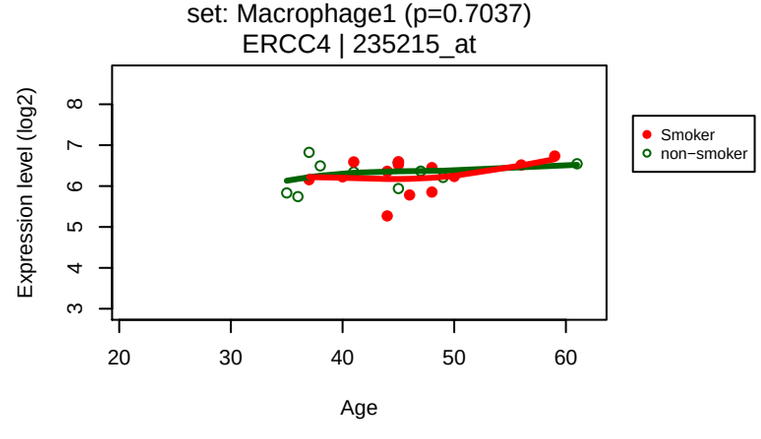

set: SmallAirway1 (p=0.2268)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker

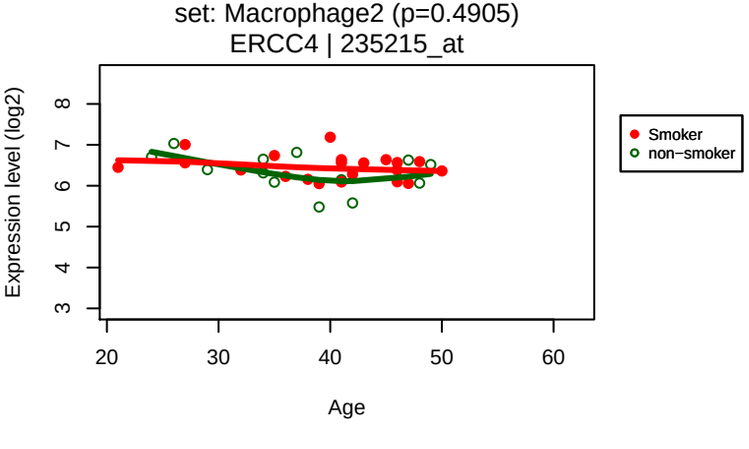

set: SmallAirway2 (p=0.0771)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker

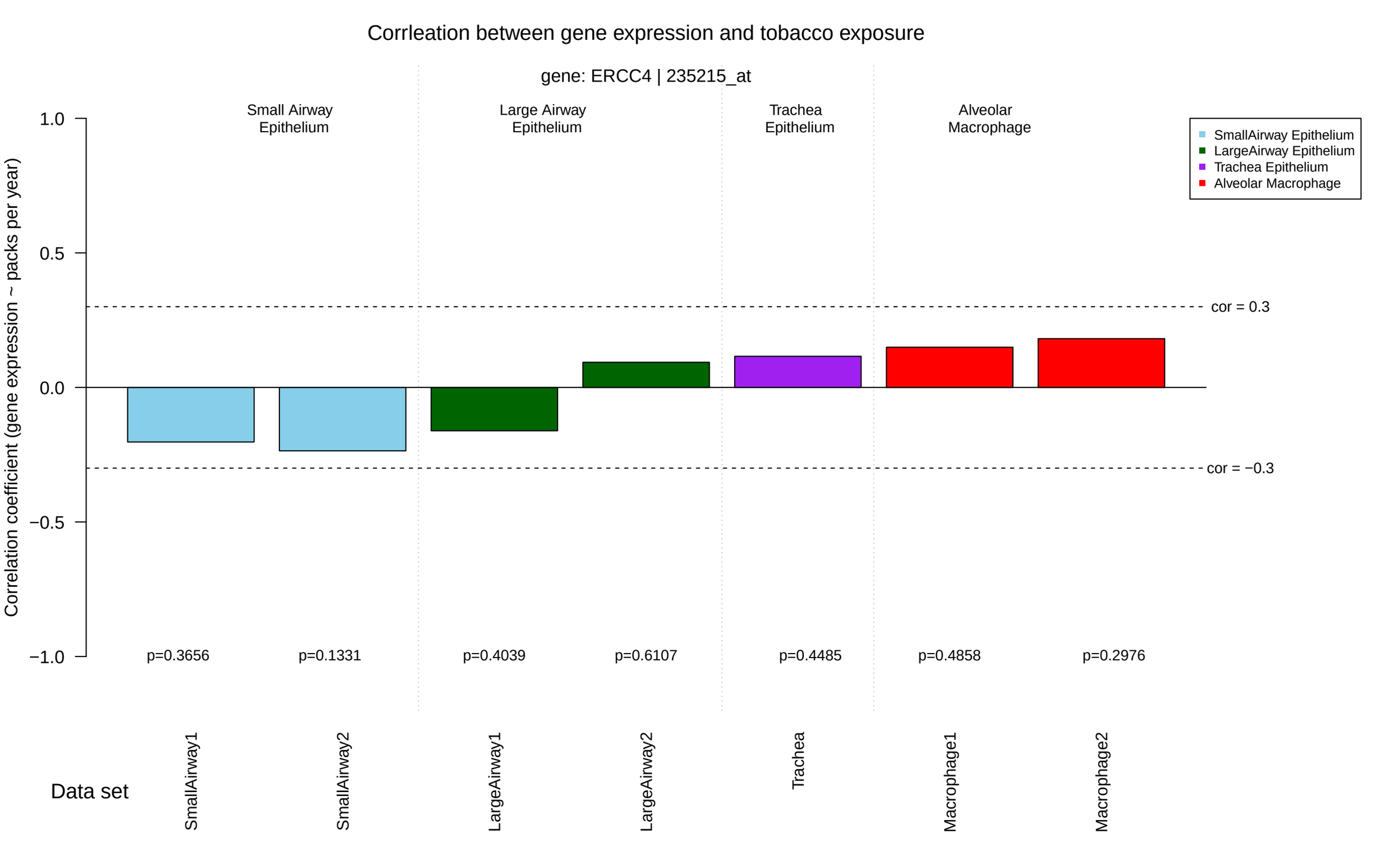

set: LargeAirway1 (p=0.0346)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4561)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9519)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7037)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4905)
ERCC4 | 235215_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERCC4 | 235215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3656
p=0.1331
p=0.4039
p=0.6107
p=0.4485
p=0.4858
p=0.2976
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2